

Research Proposal

MSc Thesis in Data Science and Business Analytics

Efficient large scale computation of sparse travel time matrices between gridded population points and points of interest using OpenStreetMap networks

by

Iana Mkrtchian, 16038592

Date: 15.03.2026

Introduction *[describe the problem, the setting, the current state of knowledge, using relevant literature and introduce the research question]*

Geospatial analysis often requires the computation of travel times between population points and points of interests (POIs), which can become quite costly and even infeasible if done for a large number of origins and destination points. Recent developments in accessibility analysis and routing algorithms provide opportunities for faster shortest path computations on large road networks. However, it is still a challenge to enable efficient scalable approaches for sparse travel time matrix construction. Therefore, this research investigates how it can be done in an efficient manner.

Research questions: *[actual question that the research is aiming to answer (given your current state of knowledge)]*

How can sparse travel-time matrices between population grids and points of interest be computed efficiently at large geographic scales using road network data?

Methodology and Techniques *[including the type of models or data science techniques that you plan to build on and use]*

This research aims to develop an approach for efficiently computing sparse travel time matrices between gridded population points and points of interests (POI) by implementing the following methods and techniques:

- Geospatial analysis using road network data derived from OpenStreetMap.
- Population grid to POI accessibility modelling.
- Spatial indexing and nearest-neighbor search techniques, such as KD trees, ball trees, R trees or geospatial trees, for efficient candidate pair filtering.
- Network based shortest path computation on large graphs
- Parallel and scalable computation techniques.

Data and/or expected results *[you should indicate the type of results you expect or possible outcomes of your research you anticipate and which data you expect to have available]*

This research will use road network data derived from OpenStreetMap, population locations that will be represented by gridded population dataset, and points of interests such as healthcare facilities that can be obtained by available geospatial datasets.

The expected outcome is an approach or technique that will enable faster computation of large-scale accessibility matrices.

Literature [*Key References*]

Boeing, G. (2017). OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65, 126–139. DOI: [10.1016/j.compenvurbsys.2017.05.004](https://doi.org/10.1016/j.compenvurbsys.2017.05.004).

Boeing, G., Higgs, C., Liu, S., Giles-Corti, B., Sallis, J. F., Cerin, E., ... & Arundel, J. (2022). Using open data and open-source software to develop spatial indicators of urban design and transport features for achieving healthy and sustainable cities. *The Lancet Global Health*, 10(6), e875-e888. DOI [https://doi.org/10.1016/S2214-109X\(22\)00072-9](https://doi.org/10.1016/S2214-109X(22)00072-9)

Boeing, G. (2025). Modeling and analyzing urban networks and amenities with OSMnx. *Geographical Analysis*, 57(4), 567–577. DOI: 10.1111/gean.70009

Delling, D., Goldberg, A. V., Nowatzyk, A., & Werneck, R. F. (2010). PHAST: Hardware-accelerated shortest path trees (Technical Report MSR-TR-2010-125). Microsoft Research.

Delling, D., & Werneck, R. F. (2013). Customizable point-of-interest queries in road networks. *Proceedings of the 21st ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, 490–493. (Also published as Microsoft Research Technical Report MSR-TR-2013-90). DOI <https://doi.org/10.1145/2525314.2525449>

Giannotti, M., Tomasiello, D. B., & Bittencourt, T. A. (2022). The bias in estimating accessibility inequalities using gravity-based metrics. *Journal of Transport Geography*, 101, 103337. DOI: [10.1016/j.jtrangeo.2022.103337](https://doi.org/10.1016/j.jtrangeo.2022.103337)

Geisberger, R., Sanders, P., Schultes, D., & Delling, D. (2008). Contraction Hierarchies: Faster and Simpler Hierarchical Routing in Road Networks. In *Workshop on Experimental Algorithms* (pp. 319-333). Springer, Berlin, Heidelberg. DOI https://doi.org/10.1007/978-3-540-68552-4_24

Higgins, C. D., DeJohn, A., Palm, M., Farber, S., Widener, M., & Miller, E. J. (2022). Calculating place-based transit accessibility: Methods, tools and algorithmic dependence. *Journal of Transport and Land Use*, 15(1), 95–116. DOI <https://doi.org/10.5198/jtlu.2022.2032>

Liu, S., Higgs, C., Arundel, J., Boeing, G., Cerdera, N., Moctezuma, D., Cerin, E., Adlakha, D., Lowe, M., & Giles-Corti, B. (2021). A generalized framework for measuring pedestrian accessibility around the world using open data. arXiv preprint arXiv:2105.08814. (Later published in Geographical Analysis). DOI <https://doi.org/10.48550/arXiv.2105.08814>

Pönkänen, M., Tenkanen, H., & Mladenović, M. (2025). Spatial accessibility and transport inequity in Finland: Open source models and perspectives from planning practice. *Computers, Environment and Urban Systems*, 116, 102218.

Weiss, D. J., Nelson, A., Vargas-Ruiz, C. A., Gligorić, K., Bavadekar, S., Gabrilovich, E., ... & Gething, P. W. (2020). Global maps of travel time to healthcare facilities. *Nature Medicine*, 26(12), 1835-1838. DOI <https://doi.org/10.1038/s41591-020-1059-1>

Ye, P., Ye, Z., Xia, J., Zhong, L., Zhang, M., Lv, L., Tu, W., Yue, Y., & Li, Q. (2024). National-scale 1-km maps of hospital travel time and hospital accessibility in China. *Scientific Data*, 11(1130). DOI <https://doi.org/10.1038/s41597-024-03981-y>

Timetable *[of activities and details of intended supervision meetings]*

1st Month

Activity	Date
Preparation and submission of thesis proposal	1.
Data acquisition and literature review	2.

2nd Month

Activity	Date
Implementation of spatial indexing and filtering methods	3
Implementation of shortest path-computation and sparse travel matrix construction	4.

3rd Month

Activity	Date
Evaluation and accessibility analysis	5.
Thesis finalization and submission	6.

Make sure you are enrolled in Canvas > MSc Data Science and Business Analytics, the website that is used for sharing information. If this is not the case contact the program director.

Also, make sure you are enrolled in the track-dependent course for uploading your thesis.

I am enrolled in the course:

Master's Thesis Data Science

☐

Master's Thesis Business Analytics

☐

Supervisor

Signature

Joaquim Gromicho

Second marker (suggestion by supervisor)

Dick den Hertog

Thesis coordinator

Signature

C.G.H. Diks